

HTML CSS & JS

*HTML Syntax and Core Concepts
(Understanding the Building Blocks of HTML)*

Objectives

- Understand HTML elements and tags.
- Learn about attributes and their usage.
- Know how to write comments in HTML.
- Be familiar with HTML entities for special characters.

HTML Elements and Tags

- **HTML Element**

- A component of an HTML document, consisting of a start tag, content, and an end tag.
- Tells the browser something about the content

- **Void Element**

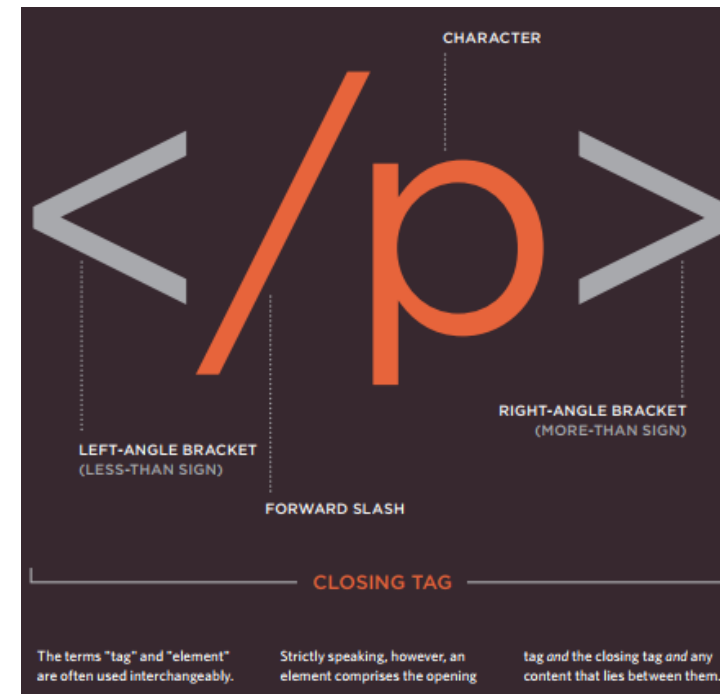
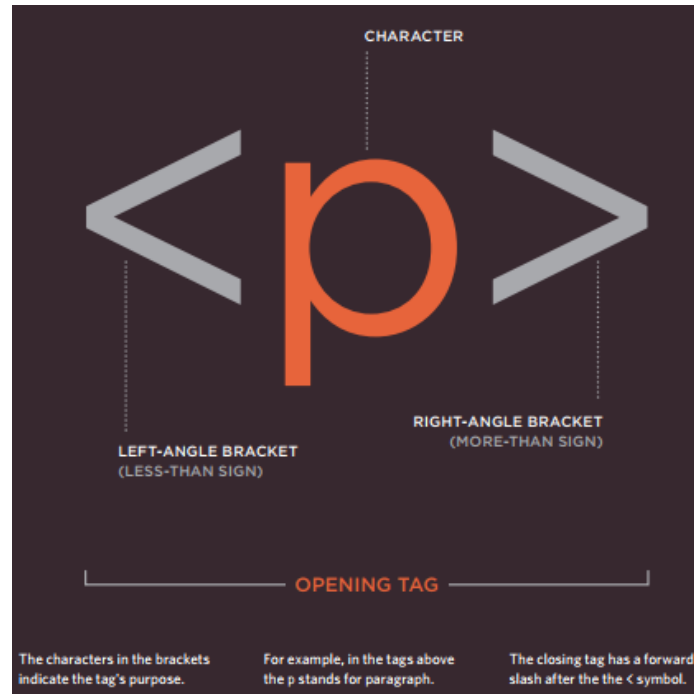
- Cannot have any child nodes (i.e. nested elements)
- **Only** has a start tag, end tags **must** not be specified
- E.g. `<br>`, `<img>`, `<meta>`, etc

- **Tags**

- They are markers
- Define the beginning (`<tag>`) and end (`</tag>`) of an element.

```
<!-- <p>: Opening tag, "This ... page": Content, </p>: Closing tag-->
<p>This is my first HTML page.</p>
```

HTML Elements and Tags



Attributes

- They provide additional information about the contents on an element
- Mostly used on certain attributes.
 - Except global attributes which appear on any element
- Placed with the start tag
- Syntax
 - `name="value"`
- [Read more here](#)

Attribute Name	Common Element associate	Description
id	Global attr	Unique identifier for an element
class	Global attr	Class name for styling or JavaScript
src	, <input>, <script>	Source URL for images or scripts
href	<a>, <link>	Hyperlink reference for anchor tags
alt	<input>, 	Alternative text for images

```
<h1 class="section-header">Hello, World</h1>
```

Attributes

- Most attributes values are either pre-defined or follow a stipulated format
- **Boolean attribute**
 - Do not require a value
 - Their presence implies true
 - E.g. *disabled, readonly, required, checked*

```
11 <button disabled>Submit</button>
```

Comments

- Used to leave notes or explanations in the code
- Not displayed in the browser
- Good practice to add comments to your code
 - Makes it easier to understand for someone else
 - Even to yourself, when you come to the code later
- Syntax

```
11      <!-- your comment goes here -->
12      <button disabled>Submit</button>
```

Nesting Elements

- Placing elements inside other elements
- Always ensure proper opening and closing of tags
- Avoid overlapping tags

```
14      <!-- Correct nesting -->
15      | <!-- <strong> is nested in <p>, and <p> is nested inside <div> -->
16  ✓    <div>
17      | <p>My name is <strong>John Doe</strong></p>
18      | </div>
```

```
14      <!-- Incorrect nesting -->
15      | <!-- <strong> is overlapping with <p> -->
16  ✓    <div>
17      | <p>My name is <strong>John Doe</p></strong>
18      | </div>
```


Character References

- Used to display reserved characters or special symbols
- Also referred to as **HTML entities**
- There are three types
 - **Named character references** – uses name string between ampersand (&) and a semicolon (;)
 - E.g. `<` – less than symbol, `©` – copyright symbol (©)
 - **Decimal numeric references** – Starts with `&#` followed by one or more base-ten integers, and ends with `;`. E.g. `<` – less than symbol.
 - **Hexadecimal numeric references** – Starts with `&#x` or `&#X` followed by one or more hex digits, and ends with `;`. E.g. `<` or `<` – less than symbol.
- [Read more here](#)

Case Sensitivity and Whitespace

- HTML tags and attributes are not **case-sensitive**, but lowercase is recommended for consistency.
- **Whitespace:** Multiple spaces and line breaks are collapsed into a single space in the browser.

Structuring Content with HTML

Text Formatting

- Headings (<h1> to <h6>)
 - Define the hierarchy and structure of content.
- Paragraphs (<p>)
 - Represent blocks of text or paragraphs.
- Line Breaks and Horizontal Rules
 - Line Break
: Inserts a line break without starting a new paragraph.
 - Horizontal Rule <hr>: Inserts a horizontal line
- Lists
 - Unordered List () and List Items (): Displays items in a bulleted list
 - Ordered List () and List Items (): Displays items in a numbered list

Structuring Content with HTML

Text Formatting

```
<ol>  
  <li>Item 1</li>  
  <li>Item 2</li>  
  <li>Item 3</li>  
</ol>
```

```
<ol>  
  <li>Item 1</li>  
  <li>Item 2</li>  
  <li>Item 3</li>  
</ol>
```

Structuring Content with HTML

Text Formatting

- Text Styles
 - **Bold Text**
 - `<strong>` : Indicates strong importance (semantic), typically renders as bold.
 - `<b>` : Stylistically draws reader's attention without implying any importance, also renders as bold
 - Advised to use CSS if bold is just for styling
 - *Italic Text*
 - `<em>`: Indicates emphasis (semantic), renders as italic
 - `<i>`: Stylistically offsets text without implying emphasis, also renders as italic
- **Note:**
 - At first `<b>` and `<i>` were purely used for presentation(styling) purposes. However, with the coming of HTML5, these elements have been defined to convey specific semantic meaning beyond styling.
- **Best Practices**
 - Use `<b>` and `<i>` for their semantic meaning
 - Use CSS for presentation: Utilize properties like `font-weight` and `font-style` respectively

Forms

- Form tag (<form>): used to present a document section containing interactive controls for submitting information.
- Form attributes:
 - **name**: name of the form. The value must not be empty
 - **action**: Specifies the URL where to send the form data. For instance, a server endpoint. This must also be specified
 - **method**: Specifies the HTTP method (GET or POST). Defines how the form will be sent to the server.
 - **enctype**: Specifies how the form data should be encoded
 - E.g. multipart/form-data, text/plain,

```
<form action="/submit-form" method="POST" enctype="multipart/form-data">  
  <!-- Form elements go here -->  
</form>
```

<input> Input Element

- Used to create interactive controls for web-based forms
 - To accept data from the data
- The most powerful and complex element in all of HTML
 - They have a number of combinations of input types and attributes
- Common <input> types:
 - **text**: Single-line input
 - **password**: Password input (value is masked)
 - **email**: Email input with built in validation
 - **number**: Numeric input
 - **date**: Date picker
 - **checkbox**: Used for multiple selections out of a set of options
 - **radio**: Used for single selection among multiple options in a radio group
 - **button**: renders a simple push button. However, <button> is now preferred for buttons

<input> Input Element

```
<form action="/submit-form" method="POST" enctype="multipart/form-data">

  <input type="text" name="username" placeholder="Enter your username">
  <input type="password" name="password" placeholder="Enter your password">
  <input type="email" name="email" placeholder="Enter your email">
  <input type="number" name="age" placeholder="Enter your age">
  <input type="date" name="birthdate">

  <input type="checkbox" id="subscribe" name="subscribe" value="yes">
  <!-- 'for' attribute refers to the id of the input tag which it is represent, thus above input tag -->
  <label for="subscribe">Subscribe to newsletter</label>

  <input type="radio" name="gender" value="male"> Male
  <input type="radio" name="gender" value="female"> Female
</form>
```


<input>Input Element

- Results of the previous form code snippet.
- It might look different on your machine after running because of the CSS which was applied.

Enter your username	Enter your password	Enter your email	Enter your age
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dd/mm/yyyy 	☐ Subscribe to newsletter	☐ Male	☐ Female
--	--	----------------------------	------------------------------

Textarea, Select and Button

- Textarea (<textarea>): For multi-line text input.
 - Notice this is not a void element as it is with input html element
- Select (<select>): with <option> for dropdown lists
- Button (<button>): For clickable buttons
 - It is now the favoured way to create a button than the button input element .

```
<textarea name="comments" rows="4" cols="50" placeholder="Enter your comments"></textarea>

<select name="country">
  <option value="mw">Malawi</option>
  <option value="za">Zambia</option>
  <option value="tz">Tanzania</option>
</select>

<button type="submit">Submit</button>
```

Form child Elements' Attributes

- **required:** ensures that filled out
- **placeholder:** Provides a hint within the input field
- **disabled:** Makes the input field non-editable
 - Its value cannot be submitted
- **readonly:** Makes input field non-editable but still submittable
- **title:** Provides additional information or tooltips for the input field when hovered
 - Provides validation hints

```
<input type="text" name="username" placeholder="Enter your username" required title="Provide a username">
<input type="text" name="readonlyField" value="Cannot edit this" readonly>
<input type="text" name="disabledField"
  value="Cannot edit or submit this" disabled title="Tooltip for the disabled input field"
>
```

Enter your username Cannot edit this Cannot edit or submit this

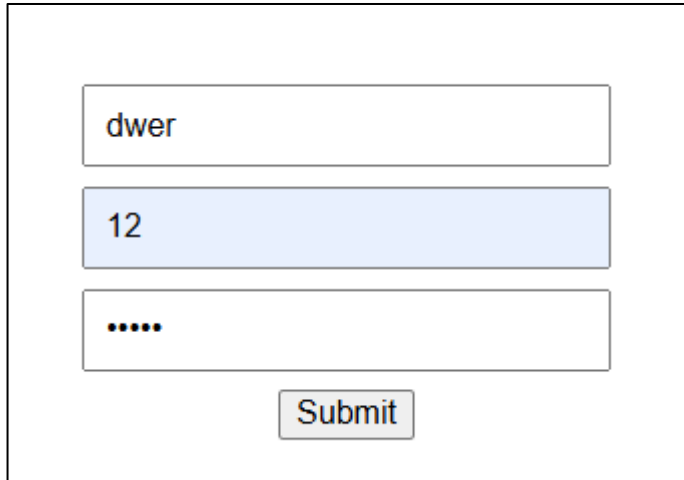
Tooltip for the disabled input field

Form Validation

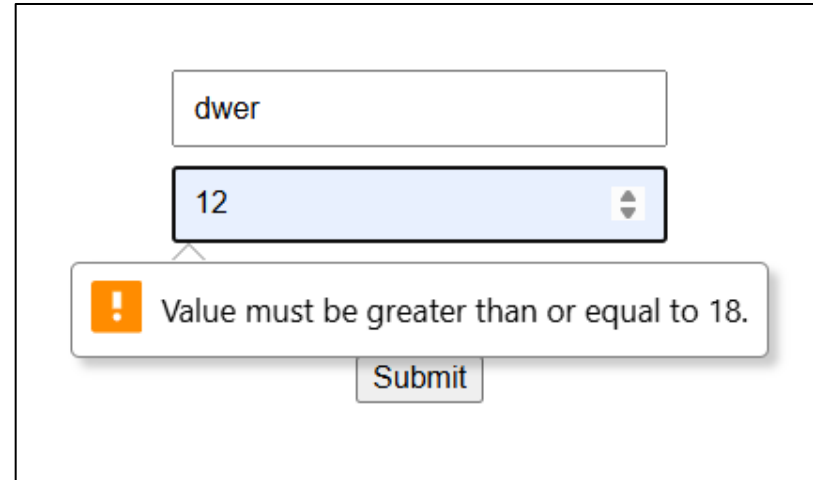
- HTML5 provides built-in validation for input fields.
 - Before HTML5, JavaScript was only used to offer validation (client-side, and server-side)
- Common Attributes and there associate input types
 - **pattern**: Specifies a regular expression for validation.
 - The input field must match in order the value to pass.
 - Used for `text`, `search`, `url`, `tel`, `email` and `password` input type
 - **min** and **max**: Defines minimum and maximum values for numeric (`number`) inputs
 - **minlength** and **maxlength**: Define minimum and maximum length (number of characters) for `text` inputs

Form Validation

```
<div class="input-attributes">  
  <input type="text" name="username" pattern="[A-Za-z]{3,}" title="Username must be at least 3 letters" placeholder="Username" required>  
  <input type="number" name="age" min="18" max="100" title="Age must be between 18 and 100" placeholder="Enter age">  
  <input type="password" name="password" minlength="8" title="Password must be at least 8 characters" placeholder="Password">  
  
  <button class="submit-button" type="submit">Submit</button>  
</div>
```



A form with three input fields and a submit button. The first field (username) contains 'dwer'. The second field (age) contains '12' and has a light blue background. The third field (password) contains eight dots. The submit button is labeled 'Submit'.



The same form as the previous one, but with a validation error. The age field now has a light blue background and a small error icon (a circle with a diagonal line) on its right side. A tooltip box is displayed below the age field, containing an orange exclamation mark icon and the text 'Value must be greater than or equal to 18.' The submit button is still labeled 'Submit'.

Note: The code snippet is a child of the form element

References

- <https://developer.mozilla.org/en-US/docs/Web/HTML>
- Duckett, J. (2011). *HTML & CSS - Design and Build Websites*. Indianapolis, USA: John Wiley & Sons, Inc.